

MPC-based Battery Thermal Management Controller for Plug-in Hybrid Electric Vehicles

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Abstract—This paper proposes a model predictive control (MPC) scheme for the battery thermal management system (BTMS) of a given plug-in hybrid electric vehicle, namely the Toyota Plug-in Prius. As temperature plays a key-role in battery life and performance, BTMS design has become a critical problem in all of the battery-based technologies. Although BTMS control design in its basic form can be well represented by a reference tracking problem, there exists only little research in the literature addressing this important control problem. Due to the importance of a prediction component in thermal systems, the idea in this paper is to design a concrete BTMS controller using the nonlinear model predictive control (NMPC) theory and examine its applicability to fill this gap. The promising simulation results indicate the prosperity of the proposed BTMS control methodology and thus pave the way for use of the model-based thermal management techniques in a wide range of upcoming battery-based devices.

I. INTRODUCTION

The growing need for environmentally-friendly power resources has drawn significant attention to clean technologies. In this regard, the automotive industry has invested in battery electric and plug-in hybrid electric vehicles (BEV/PHEVs) worldwide, leading to vast studies on battery systems as a major power source for cars. These studies show that amongst the main determinants of the battery life and performance, its temperature plays a key role, delineating the importance of a sound BTMS controller.

The literature has proposed different approaches to address the challenging problem of BTMS design; however, there are very few examples including a control methodology. Among these examples lies an electric vehicle (EV) BTMS controller developed by Bauer et al. [1], applying the Pontryagin's maximum principle (PMP) for the offline optimization of battery life expectancy and energy usage. Their design successfully maintained the battery temperature at the desired reference point. As for another example of optimal control applications, Tao et al. [2] suggested a BTMS controller for hybrid electric vehicles (HEVs) based on stabilizing the battery core temperature. They implemented this idea using a linear optimal controller and a nonlinear back-stepping controller, and reported improvements in terms of the power conservation and reference tracking. Moreover, Tao et al [3] designed a BTMS controller for HEVs based on

feedback optimal-control. This controller tracked the battery-core reference temperature while significantly reducing the power consumption. Further, Masoudi et al [4] demonstrated the application of a Dynamic Programming based optimal controller in BTMS of EVs, where the desirable temperature was tracked and the power consumption was observed to be less than that of a benchmark PID. Rule-based controllers have been also deployed in battery-related applications, as done by Pham et al. [5] for energy management of heavy duty hybrid electric trucks. Their strategy improved the fuel economy while taking into account the battery life.

It is observed that BTMS can be implemented by means of more practical control approaches as well; Shin et al. [6], [7] developed a real-time BTMS for EVs by regulating the physical battery-variables. Their simulation results showed salient performance improvements.

Motivated by the above-mentioned ideas and concerned by the existing drawbacks, this study designs a BTMS controller for a given PHEV, namely the Toyota Prius, as an optimal control tracking problem. As incorporating some prediction of the system's behaviour into the control strategy can lead to significant performance improvement in thermal systems, and also because of thermal systems' highly nonlinear nature, NMPC is chosen as the proper control approach. It is worth mentioning that due to its significant constraint handling capability [8] [9], NMPC can also very well address the state and input constraints, as critical aspects of the BTMS problem. The NMPC controller is formulated as an optimal control problem (OCP) and then turned into a nonlinear programming (NLP) problem which is on-line implementable [10], and therefore suited to the automotive industry qualifications [11] [12]. The application of this NMPC control approach to the baseline PHEV is examined by several simulations with different optimization algorithms and over a variety of driving scenarios. Simulation results show improvements compared to the current BTMS, which is comprised of a rule-based controller actuating an on-off fan.

The remainder of this paper is presented as follows: Section II explains the mathematical basis of the applied NMPC approach. Section III states the BTMS problem formulation and model components, followed by an introduction of the objectives and constraints associated with this case study.

Section IV shows and discusses the simulation results for different driving scenarios, and finally Section V accentuates the conclusions of this research.

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II. CONTROLLER DESIGN

In this section the applied control techniques are explained. First the NMPC-based controller is described explicitly. Afterwards, the benchmark methodology, namely rule-based approach, is explained.

A. NMPC-based Controller

To formulate the reference tracking problem in MPC format, at each time step a finite horizon OCP as follows is solved:

$$\begin{aligned} & \underset{u \in \mathcal{U}}{\text{minimize}} && J(x_0, u) = \sum_{i=1}^N \ell(x^u(i), u(i)), \\ & \text{subject to} && x^u(i+1) = f(x^u(i), u(i)), \\ & \text{and} && x^u(i) \in \mathcal{X}, \end{aligned} \quad (1)$$

where N is the prediction horizon, and x and u represent the state and input respectively. Index u in x^u shows the predicted input, and x_0 stands for the state value at each time-step. $\mathcal{X}$ is the state admissible region, and $\mathcal{U}$ is that of the input. Finally, f is the dynamic of the system, modelled as an equality constraint to be satisfied.

The above-mentioned problem is reformulated using the so-called single shooting procedure, where the control law sequence is considered as the optimizer.

Now equation 1 can be written as a standard NLP as follows, so that the NLP solvers could be applied to it [13]:

$$\begin{aligned} & \underset{u \in \mathcal{U}}{\text{minimize}} && V(x_0, u) = \sum_{i=1}^N \ell(x^u(i), u(i)), \\ & \text{subject to} && G(u) = G(x^u(i) - f(x^u(i), u(i))) \leq 0. \end{aligned} \quad (2)$$

As it is observed, there is no equality constraint, and as the state bounds are written in terms of the input, there are $N \cdot n$ inequality constraints. The optimization problem is now solved, and the first element of the optimal control sequence is applied to the system model.

There are different methods to solve this problem. In this research, however, the MATLAB function `fmincon` is applied. `Fmincon` offers a variety of features, and here the interior-point solver is chosen to solve the problem, where inequality constraints are turned into equality constraints using slack variables. The KKT conditions are then applied to the problem, and the resulting system of equations is solved using either direct step or conjugate gradient methods.

B. Rule-based Approach

Rule-based approaches are commonly used in thermal management systems, including that of the base-line platform, where the actuators get started once temperature exceeds 35°C [14] [15]. This rule is identified based on a rule of thumb stating that the temperature should not exceed 40°C at any point within the module.

This is the default algorithm in the vehicle; thereby, its application as benchmark gives insight regarding the performance of the designed controller.

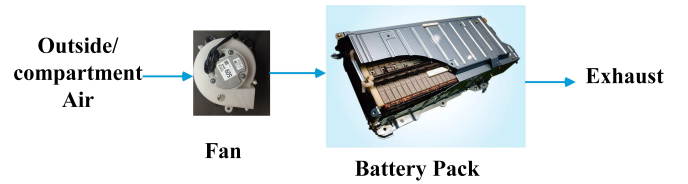


Fig. 1. Passive Cooling System Layout. The battery could be cooled down with either outside or cabin air. The latter is the case in this study.

III. BTMS PROBLEM FORMULATION

Lithium-ion batteries perform best in a certain temperature range [14], and the BTMSs are designed to maintain the battery temperature within that range. The vehicle in this case study, namely the Toyota Plug-in Prius 2015, has a passive air-based battery-cooling system, where there exist two fans providing the battery with the compartment air to cool it down. The fans are activated by the Electronic Control Unit (ECU) once the temperature exceeds some desirable level, obtained by the system operating condition [16]. The rules governing the ECU in this regard can be replaced by NMPC controller, which is formulated to calculate the control law, required air velocity in this case, such that the reference temperature is tracked. The rest of this section discusses different components of the proposed BTMS controller in more detail.

A. BTMS Layout

Figure 1 shows the architecture of a fan-based BTMS, similar to that in the base-line vehicle. This BTMS is passive and has parallel flow. The fans are activated based on the default rules, in that they start working once temperature exceeds the reference point. The fan sucks air from the passenger compartment into the battery pack, and the warmed air is exhausted.

B. Battery High-fidelity Model

Model accuracy is a key component in MPC's performance, and the applied model should be capable of reflecting different aspects of the battery thermal behaviour. The vehicle battery pack is comprised of modules; thus, a battery module is modelled, and the results are extended to the pack. Here, the considered module model is a combination of the ones offered by [14] and [15], with some modifications. As it will be explained later, the first model is applied for cases without thermal management, namely with off fan, while the latter case represents the battery behaviour exposed to the fan air.

As for the controller purposes, the so-called external inputs to the system are the current load and cabin temperature, and the air velocity of the cooling fan is the control input. The output (state) is the battery temperature. The following considerations are made before deriving the thermal model of the battery [17]:

Consideration 1: Battery cell is considered as a lumped mass, with the same core and surface temperatures.

Consideration 2: The heat transfer to the battery cell is a combination of conduction and convection. Convection heat transfer could be either forced, representing the presence of fan, or natural, for when there is no thermal management.

Consideration 3: The battery cells are considered as a set of staggered tubes [18] constructing each battery module.

Consideration 4: The only heat generation form is the Joule heating caused by battery internal resistance.

Consideration 5: The cooling system has parallel flow; therefore, only one module can be modelled and simulated, and the results are extended to the whole battery pack

As suggested in Consideration 2, the applied battery model is comprised of two scenarios, with and without the thermal management. In the first case, any change in module temperature can be because of either the heat generation or the heat transferred with the air flowing around the module. As such, one could write

$$\dot{T} = \frac{\dot{Q}_{gen} - \dot{Q}_{tf}}{m_{bat}C_{p,bat}}, \quad (3)$$

where $\dot{Q}_{gen}$ and $\dot{Q}_{tf}$ are rates of the heat generation and heat transfer respectively, and m_{bat} and $C_{p,bat}$ stand for module mass and thermal capacity.

Now $\dot{Q}_{gen}$ and $\dot{Q}_{tf}$ should be calculated; due to Consideration 4, the module heat-generation becomes

$$\dot{Q}_{gen} = NR_{bat}I_{bat}^2, \quad (4)$$

with N being the number of cells per module, and I_{bat} representing the current of each cell.

Based on Consideration 3,

$$\dot{Q}_{tf} = N\pi h d \Delta T L, \quad (5)$$

where d is the cell diameter, L is the cell length, and ΔT is the log mean temperature difference defined as

$$\Delta T = \frac{(T - T_{in}) - (T - T_{out})}{\ln\left(\frac{(T - T_{in})}{(T - T_{out})}\right)}. \quad (6)$$

Finally, h , or heat transfer coefficient, can be obtained using

$$h = \frac{Nuk}{d}, \quad (7)$$

where k is air thermal conductivity and Nu is the Nusselt number, or

$$Nu = CRe_{max}^m Pr^{0.36} \left(\frac{Pr}{Pr_s}\right)^{1/4}. \quad (8)$$

Here, Pr is the Prandtl number, and Re_{max} is the Reynolds number at maximum air velocity, both of which are calculated at the film temperature,

$$T_f = (T + T_{in})/2. \quad (9)$$

C and m are constants evaluated at Re_{max} , as shown in Table I. Here, it has been assumed that the maximum velocity is equal to the average velocity provided by the controller.

TABLE I
CONSTANTS OF EQUATION 8 FOR STAGGERED FLOW [18]

Reynolds	C	m
$10 - 10^2$	0.90	0.40
$10^2 - 10^3$		Isolated cylinder approximation
$10^3 - 2 \times 10^5$	0.40	0.60
$2 \times 10^5 - 2 \times 10^6$	0.022	0.84

When there is not any air flow-rate, the battery temperature is found as follows [14]:

$$\dot{T} = \frac{\dot{Q}_{gen} - (T - T_{in})/R_{th}}{m_{bat}C_{p,bat}}, \quad (10)$$

where R_{th} is the equivalent thermal resistance associated with conduction and natural convection. It is a constant value equal to that applied in the Autonomie software.

To be implemented, Equations 3 and 10 are discretized using proper step-lengths.

C. Optimization Objectives

As previously mentioned, the NMPC-based controller achieves tracking by calculating the required air velocity; however, there should also be some considerations regarding keeping this value low. To fulfil both of these objectives, the instantaneous cost function can be expressed as

$$\ell(k, x, u) = w_1(\theta(k) - \theta_{ref})^2 + w_2 V(k)^2, \quad (11)$$

where the first term in the right hand side (RHS) accounts for minimizing the error between the actual and reference temperature, and the second RHS term is in charge of the control effort minimization. The weighting factors w_1 and w_2 facilitate a trade off between the two objectives.

D. Physical Constraints

The feasible regions are precisely identified for the existing BTMS. Here, the state feasible region is obtained by the operating condition, which can be considered as $\mathcal{X} = [25, 40]$ (temperature in terms of $^{\circ}\text{C}$). Moreover, the control input is bounded as it represents the air velocity in an actual fan; therefore, the control feasible region can be stated as $\mathcal{U} = [0, 24]$ (air velocity in terms of m/s).

E. Control-oriented model

For the purpose of simulation, a control-oriented model is applied within the controller algorithm, which has to be simple yet accurate. In the first case, namely forced convection, many of the parameters are modelled as maps; therefore, this model is simplified to a constant-parameter control-oriented model as follows:

$$T_o = T - (T - T_{in}) \exp\left(-\frac{A_3}{V}\right), \quad (12)$$

$$\dot{T} = 3A_1 I^2 - (A_2 V^m) \frac{(T - T_{in}) - (T - T_o)}{\ln((T - T_{in})/(T - T_o))}.$$

, where parameters A_1 , A_2 , and A_3 are obtained through least square-based cross-validation with the main model, over a

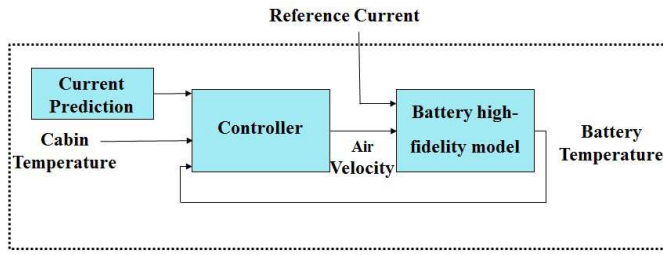


Fig. 2. MIL Simulation Set-up

combination of cycles¹. As for the natural convection case, the same model was applied as the control-oriented model as the parameters were all constant.

IV. SIMULATION RESULTS

This section investigates the performance of the designed controller over different driving scenarios², specially those with potential challenge for BTMS. The results are then compared with those of the on-off fan, which is the default in the existing design [19]. Finally, some general guidelines are offered regarding battery thermal management control system design providing insight for future research in this regard.

A. Simulation Procedure

The set-up for the conducted model-in-the-loop (MIL) simulations is shown in Figure 2, comprising of the battery model, controller, and a subsystem that measures the load-current. Here, the load-current values are calculated based on the driving cycle being driven and are assumed to be available for measurement over the prediction horizon³.

In this problem, the external input, namely the load-current, is the main factor affecting the system's response. Thus, driving scenarios are chosen such that they include different types of input behaviours; UDDS cycle is chosen as a city cycle. The aggressive US06 cycle is also applied, which is important as variable gradients of the external input could dramatically affect the behaviour of the control system. Finally, a highly variable road grade case, namely Calgary-Vancouver road driven at a constant speed of 90 km/h, is considered. Road-grade is a physical phenomenon having a great impact on the load-current, and specifically it is investigated here because of the slow nature of the thermal systems. By prediction the system is meant to be made responsive to sudden input changes.

One of the controller parameters is the so-called actuation-temperature, which is the temperature at which the NMPC-based controller is activated; in all of the simulations, it is assumed to be 25 °C, to facilitate enough time for foreseeing

¹A combination of UDDS and US06 driving cycles as well as Calgary-Vancouver road driven at a constant speed of 90 km/h was applied for cross-validation.

²The velocity in the driving cycle is turned to load current, or controller input, using the Autonomie software.

³Technologies such as radar, V2V, and V2I are examples of how the power demand could be predicted.

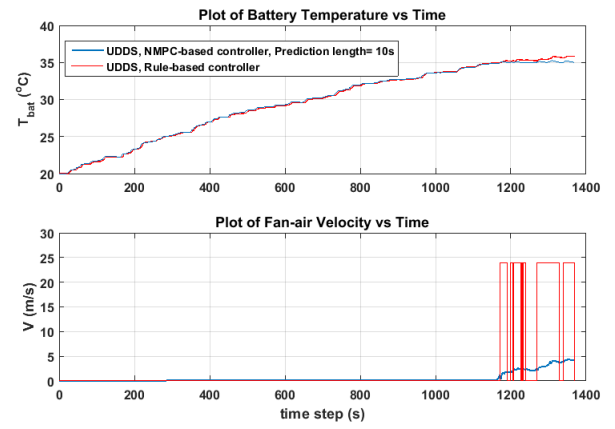


Fig. 3. NMPC-based controller versus the rule-based one over UDDS cycle.

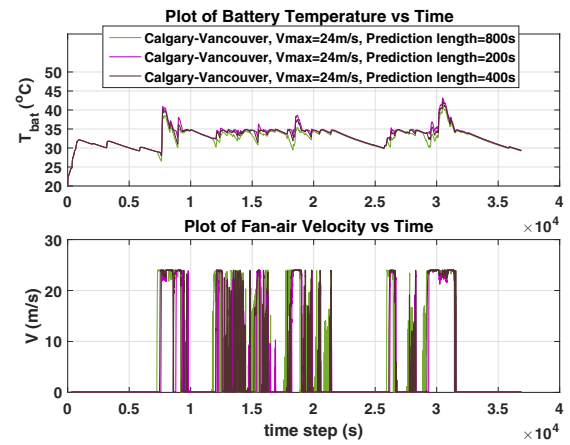


Fig. 4. MIL Results for NMPC: Calgary-Vancouver route results with different prediction lengths.

the system's evolution. In each of the simulation cases, the weighting factors of the objective function, w_1 and w_2 , are tuned relatively. The reference temperature is always $T_{ref} = 35^\circ\text{C}$. Moreover, T_{in} is considered constant and equal to 20°C , which is fair specially with the on AC.

B. Results

The first studied scenario is the city cycle, namely UDDS driving cycle. The results of the devised predictive controller, with a prediction length of 10s, as well as those of the rule-based one are shown in Figure 3.

It is observed that in such a short period of time, replacing the default rules with the predictive algorithm significantly reduces the required control input, up to almost 80%. Moreover, despite the short duration of this cycle, the temperature profile is slightly improved.

Figure 4 shows the simulation results of the highly-variable road-grade case, for different prediction horizons. High prediction lengths are acceptable in this case as the road grade ahead could be approximated based on the GPS information and road maps.

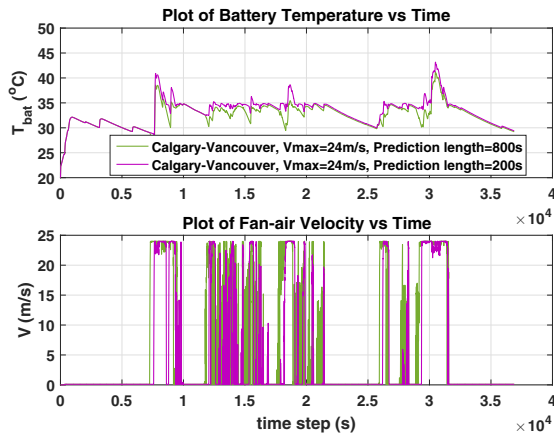


Fig. 5. MIL Results for NMPC: more detailed view of two of the cases in Figure 4

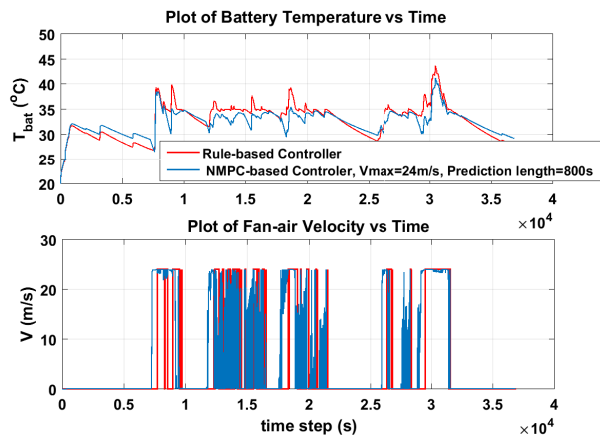


Fig. 6. NMPC performance versus rule-based controller for Calgary-Vancouver road

Based on these results, it turns out that the general grade-trend is necessary yet its precise value at every single point is not required. In one of the cases, the input upper bound has been increased to 100 m/s , which slightly shifts the temperature profile down. To facilitate an easier comparison, a more detailed view of two of the cases in Figure 4, namely with 200 s and 800 s prediction horizons, is shown in Figure 5.

As observed, the temperature profile associated with the longer prediction horizon falls below that with the lower prediction, which demonstrates the impact of predictive thermal management. This is the case without imposing significantly more value of air velocity, showing that prediction facilitates wiser application of the actuators.

Performance of the NMPC controller with 800s-prediction length versus that of the rule-based is shown in Figure 6. It is observed that the NMPC approach results in a smoother temperature profile, with less deviations from the reference temperature. Furthermore, at higher temperatures, the impact of the predictive methodology stands out.

Finally, four subsequent US06 cycles are attached to provide a longer aggressive cycle, over which the NMPC

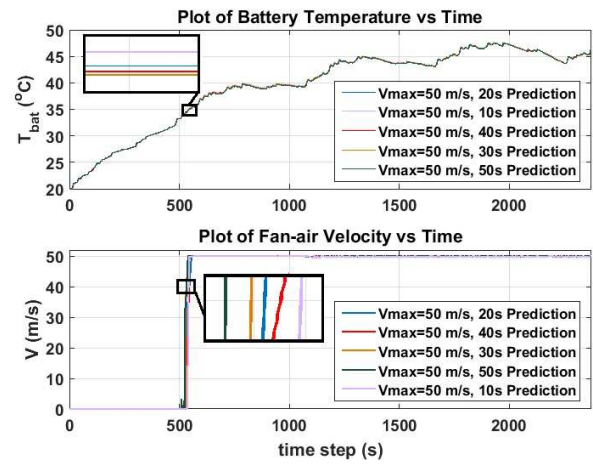


Fig. 7. MIL Result: four subsequent US06 Single Shooting Interior-point Solver. Results for realistic prediction lengths

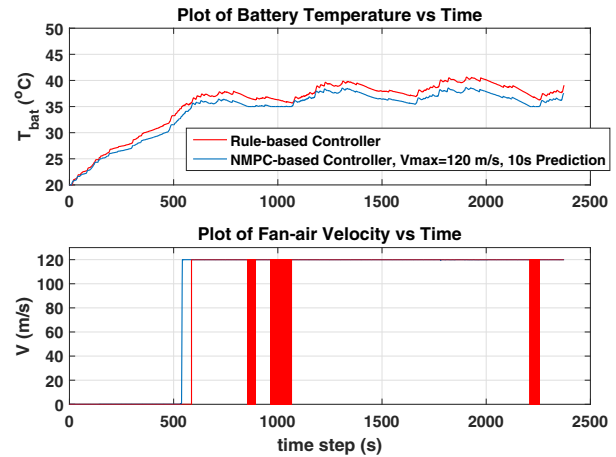


Fig. 8. MIL Result: NMPC performance versus rule-based controller for four subsequent US06 cycle

performance is examined; the results are illustrated in Figure 7 for different prediction horizons. The maximum velocity in this case is assumed to be 50 m/s rather than 24 m/s . This is done for facilitating some space for discussion; nevertheless, as it will be shown this takes further upsizing the actuator.

The maximum realistic prediction horizon considered in this case is 50 s , which is obtained by using the 1000 m prediction of V2V/V2I technologies, provided the vehicle is driven at the velocity of 20 m/s . Changing the prediction horizon in this case has not significantly improved the controller performance. Indeed, the order of predictions must be higher to show significant difference. As for the aggressive cycles, the predictive controller is insensitive to the real-life prediction ranges.

A comparison between the rule-based and NMPC controllers is demonstrated in Figure 8; the upper bound on the input has been increased to 120 m/s to keep the temperature below 40 $^{\circ}C$. This necessitates upsizing the actuator, which should be investigated in its own sense.

C. Discussion

Based on the observed simulation results, some general guidelines can be offered regarding the BTMS controller modification. When it comes to highly-variable road-grades, predictive approaches result in a better performance and are more cost-effective. As for city-driving, although the rule-based performs well, predictive approaches perform better, mainly in that they save a great amount of energy. To take care of the aggressive cycles, one should use the maximum capability of the BTMS. As such, keeping the fan on all the time strategy is recommended. This, however could be subject to change based on the cycle duration and current gradient.

V. CONCLUSION

This research demonstrates potential of NMPC in BTMS controller design for PHEVs, along with further investigating the requirements of a proper BTMS controller. The devised algorithm improves the BTMS control performance in the base-line PHEV, namely the Toyota Prius, as proven by simulation results over a variety of driving cycles. It is observed that the BTMS algorithm had better be switched to NMPC, specifically in case of highly variable road grades.

This paper takes the initiative as introducing NMPC for BTMS control problem. There exist, however, various relevant areas to be considered in the future; importance of prediction necessitates taking advantage of approaches such as trip planning to integrate route data into the control strategy. Also, the impact of temperature on battery end-of-life motivates us to define an end-of-life-wise cost-function, which would in turn help decreasing the battery replacement costs. Another interesting area is simultaneous thermal management and component-sizing of the thermal actuators, which is similar to the existing literature on supervisory control and component sizing, such as [20] [21]. Moreover, as the system is subject to many uncertainties, most importantly those caused by estimating the future power demand, studying the system robustness is a very important step before actual implementation of this mechanism on the vehicles. Another important issue could be the stability of the system which should be studied further [22].

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